Backpaper Exam
Representation Theory of Finite Groups
BMath-IIIrd year

Time: 3 hours

Max marks: 100

Answer all questions. No marks will be awarded in absence of proper justification.

Notations are as used in class. All groups are finite, and all representations are finite dimensional over $\mathbb{C}$.

1. (a) Prove that if $\phi : G \to GL(V)$ is a unitary representation, then ϕ is either irreducible or decomposable.
(b) Prove that every representation is equivalent to a unitary representation.
(c) State and prove Maschke's theorem for representations of a finite group. (6+6+8)

2. (a) Define symmetric k-tensors $\mathrm{Sym}^k(W)$ of a vector space W. Given a basis $(w_1, \ldots, w_n)$ of W, determine a basis of $\mathrm{Sym}^k(W)$.
(b) Let V be the standard representation of the symmetric group S_3, the irreducible representation of degree 2. Find a basis for V.
(c) Show that for $\sigma = (123)$ and $\tau = (12)$,

$$\chi_{\mathrm{Sym}^k(V)}(1) = k + 1$$

$$\chi_{\mathrm{Sym}^k(V)}(\sigma) = \begin{cases} 1 & k \equiv 0 \bmod 3 \\ -1 & k \equiv 1 \bmod 3 \\ 0 & k \equiv 2 \bmod 3 \end{cases}$$

$$\chi_{\mathrm{Sym}^k(V)}(\tau) = \begin{cases} 1 & k \equiv 0 \bmod 2 \\ 0 & k \equiv 1 \bmod 2 \end{cases}.$$

(6+2+12)

3. Compute the character table of A_4. Write down the irreducible decomposition of the induced representations of all the irreducible representations of A_4, as representations of S_4. (10+10)

4. (a) Define algebraic integers. Prove that the characters of a group are algebraic integers.
(b) Show that all characters of the symmetric group S_n are integers, for all $n \geq 2$.
(c) Show that if χ is the character of an irreducible representation of S_n, not equivalent to the sign representation, then

$$\sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\chi(\sigma) = 0.$$

(4+12+4)

5. (a) Prove that the number of real irreducible characters of G equals the number of real conjugacy classes of G.
(b) Deduce that G is of odd order if and only if G does not have any non-trivial real irreducible characters. (12+8)